

Spike-interaction asymptotics for the minimal cross-feeding model: how to build the reduced dynamics, and what it says about arrest

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Abstract

A programme for reducing the three-component cross-feeding PDE to an ODE for the positions of the population spikes, in the regime $D_N \ll D_I, D_R$. The reduction has four steps: an outer problem in which each spike is a point sink of I and a point source of R ; an inner problem which — because growth is linear in N — is a *linear Schrödinger* eigenvalue problem in a well the spike digs itself, rather than the algebraic balance familiar from Gray–Scott; a self-consistency condition fixing the spike masses; and a solvability condition giving the drift law $\dot{x}_j \simeq x_0^2 \partial_x \mu_{\text{out}}(x_j)$. Every step that can be checked against the arrested 29-spike state at $p = 1$ is checked here, and the outer problem and the inner virial identity both hold. The frozen-strength pair force does not: it predicts an attractive interaction with an e -folding time of 7.6×10^3 for a run that sat still to $t = 10^8$. That discrepancy localises the physics of arrest in the configuration-dependence of the spike masses, and suggests a cheap decisive numerical experiment.

1 The model and the state we are trying to explain

With $k = 0$ and unit energy weights the minimal model in one spatial dimension is

$$\partial_t N = D_N N_{xx} + N \mu(I, R), \quad \mu(I, R) \equiv (1 - l)cI + dR - m, \quad (1)$$

$$\partial_t I = D_I I_{xx} + K - rI - cNI, \quad (2)$$

$$\partial_t R = D_R R_{xx} + lcNI - rR - dNR, \quad (3)$$

on a ring of circumference L , with $D_R = p$ the scan parameter. Throughout, numbers come from `run4_arrest_check/state1.jld2`: the arrested 29-spike state at $K = 15$, $l = 0.999$, $m = c = d = r = 1$, $D_N = 10^{-6}$, $D_I = 1$, $p = 1$. Table 1 collects what that state actually looks like; all of it is reproduced by `verify_spikes.jl` in this directory.

quantity	value	
domain, spikes	$L = 35.4144$, $k = 29$	spacing $d = 1.2212$
spike amplitude, mass	$A = 255.3$, $S = 14.2598$	identical across spikes to 6 d.p.
spike width	$x_0 = 0.02204$	2nd moment; $= 0.02228$ from S/A
compare	$\sqrt{D_N/m} = 10^{-3}$	$x_0/\sqrt{D_N/m} = 22$
μ at spike core	$+1.856 \times 10^{-3}$	-0.1443 at mid-gap
I	1.0270 at core	3.1630 at mid-gap
R	1.00083 at core	0.85254 at mid-gap
resource ranges	$\ell_I = \sqrt{D_I/r} = 1$	$\ell_R = \sqrt{p/r} = 1$

Table 1: The arrested state. Note R is *elevated* at the spike and I *depleted*: R is the activator (self-produced food), I the inhibitor (the thing competed for).

Three lengths matter, and they order as

$$x_0 = 0.022 \ll d = 1.22 \sim \ell_I = \ell_R = 1. \quad (4)$$

This is the *semi-strong* interaction regime: spikes are narrow, but the fields that couple them vary on the scale of the spacing itself. In particular $d/\ell \approx 1.2$, so neighbours are *not* exponentially far apart, and $e^{-d/\ell} \approx 0.3$ is not small. Whatever makes the coarsening slow here, it is not the smallness of $e^{-d/\ell}$.

Remark. $x_0 = 22\sqrt{D_N/m}$. The naive inner scale is wrong by a factor of 22, and the reason is structural: (1) is *linear* in N , so there is no algebraic balance $D_N N'' \sim N^2$ to set a sech^2 width. The localisation is a linear bound state, and its width is set by the curvature of the well, which lives on the outer scale. This is the one place where the calculation genuinely differs from Gray–Scott, and it makes the inner problem *easier*, not harder.

2 Step 1: the outer problem — spikes as point sources

Away from the spikes N is negligible ($\sim 10^{-63}$ in the gaps of the measured state), so (2)–(3) linearise:

$$D_I I_{xx} - rI = -K, \quad D_R R_{xx} - rR = 0, \quad (5)$$

punctured at each x_j by jump conditions obtained by integrating across the spike. Writing σ_j^I, σ_j^R for the signed point-source strengths,

$$D_I [I_x]_{x_j} = -\sigma_j^I = \int_{\text{spike}} (cNI - K + rI) dx, \quad D_R [R_x]_{x_j} = -\sigma_j^R = -\int_{\text{spike}} (lcNI - dNR - rR) dx. \quad (6)$$

Measured on the state, over a window of ± 0.085 around a core, both hold to four figures:

$$\sigma^I = -12.926, \quad \sigma^R = +0.93045, \quad \text{ratio (jump)/(integral)} = 1.0000 \text{ for both.}$$

The linear terms in (6) are *not* optional: over that window $\int (K - rI) = 2.44$ against an uptake of 15.36, so dropping them costs 16%. This is a symptom of x_0/ℓ not being very small; see §9.

On the ring the solution of (5) is a superposition of Green's functions,

$$g_\ell(x) = \frac{\ell}{2D} \frac{\cosh((L/2 - |x|)/\ell)}{\sinh(L/2\ell)}, \quad Dg_\ell'' - rg_\ell = -\delta(x), \quad (7)$$

so that

$$I(x) = \frac{K}{r} + \sum_k \sigma_k^I g_{\ell_I}(x - x_k), \quad R(x) = \sum_k \sigma_k^R g_{\ell_R}(x - x_k). \quad (8)$$

This is exact up to the width of the spikes, and it is very accurate: fitting $I - K/r$ and R over one gap to a single cosh with the *predicted* $\ell_I = \ell_R = 1$, with only the amplitude free, gives relative residuals of 4.4×10^{-4} for both fields. The outer half of the reduction is essentially exact for this state.

3 Step 2: the inner problem — a self-dug potential well

Zoom in on one spike. On the inner scale I and R are quasi-static and (1) reads

$$D_N N_s'' + \mu(x) N_s = 0, \quad (9)$$

a zero-energy Schrödinger bound state in the potential μ . Equivalently, $-D_N \psi'' + V\psi = E\psi$ with $V = -\mu$ and $E = 0$. Two consequences.

The virial identity. Multiplying (9) by N_s and integrating over a period,

$$\boxed{D_N \int (\partial_x N_s)^2 dx = \int \mu N_s^2 dx} \quad (10)$$

exactly, for any spike steady state, whatever its shape. On the measured state the two sides are 2.5287 and 2.6253, a ratio of 0.963 — which for a quantity assembled from a numerical second derivative of a field spanning 65 orders of magnitude is a clean confirmation. This is the cheapest possible test that a candidate state really is a steady state of (1), and it is worth adding to the arrest-check battery: it is shape-free, so it cannot be satisfied by a grid-pinned artefact.

Self-consistency fixes the mass. Approximating the well harmonically, $\mu \approx \mu_0 - \frac{1}{2}\kappa \xi^2$ with $\xi = x - x_j$, the ground state is a Gaussian $N_s \propto e^{-\xi^2/2x_0^2}$ with

$$x_0^2 = \sqrt{2D_N/\kappa}, \quad \mu_0 = \sqrt{D_N\kappa/2} \quad \implies \quad \mu_0 x_0^2 = D_N. \quad (11)$$

The second relation is the self-consistency condition: *the growth rate at the spike core must sit exactly at the ground-state energy of the spike's own well*. Since the well is dug by the spike, this is what determines the mass S_j . The corollary $\mu_0 x_0^2 = D_N$ is parameter-free and testable: measured, $\mu_0 x_0^2 = 9.017 \times 10^{-7}$ against $D_N = 10^{-6}$, a ratio of 0.90. The residual 10% is the non-harmonicity of the well, discussed next.

What the well actually looks like. The point-source structure of (8) means both I and R have a *cusp* at the spike, so on the inner scale $\mu \approx \mu_0 - \alpha|\xi| - \frac{1}{2}\kappa\xi^2$ with $\alpha = -[(1-l)cI_x + dR_x]$ evaluated just outside the spike. Measured: $\alpha = 0.463$, of which the R part is -0.4696 and the I part only $+0.0065$ — the tent is dug by the byproduct. If one takes that tent as sharp, the ground state is an Airy function with

$$x_0 \sim (D_N/\alpha)^{1/3} = 0.0129, \quad \mu_0 = |a'_1| (D_N\alpha^2)^{1/3} = 6.10 \times 10^{-3}, \quad (12)$$

against measured 0.0220 and 1.86×10^{-3} : too narrow by $1.7\times$ and too deep by $3.3\times$. The reason is visible in the numbers — the source is not narrow compared with the well it digs, $x_0/(D_N/\alpha)^{1/3} = 1.7$, so the tent is substantially rounded by the spike's own finite width. The honest inner problem is therefore the coupled system

$$D_N N_s'' + [(1-l)cI + dR - m]N_s = 0, \quad D_I I'' = cNI, \quad D_R R'' = (dR - lcI)N, \quad (13)$$

on $\xi \in (-\infty, \infty)$, with $N_s \rightarrow 0$ and I, R matching the outer slopes $\sigma^{I,R}/2D$ as $|\xi| \rightarrow \infty$. That is a small two-point BVP with one eigenvalue-like unknown (the mass), solvable by Newton in seconds. It is worth doing properly rather than harmonically, because it is what supplies the $O(1)$ constants everywhere downstream.

4 Step 3: closing the loop — the k -spike quasi-equilibria

Steps 1 and 2 close on each other. Given positions $\{x_j\}$: (8) gives the local field values (I_j, R_j) in terms of the strengths, (13) gives the strengths in terms of (I_j, R_j) , and the fixed point is k algebraic equations for k masses. For equally spaced spikes symmetry collapses this to a single scalar equation, so the exact k -spike state is available in closed form for every k , at the cost of one Newton solve on one period — no PDE integration, and no restriction to k that the dynamics happens to visit. This is the object whose Floquet spectrum gives the stable wavenumber band.

5 Step 4: motion — the solvability condition

Now let the environment be slightly asymmetric. Write $\mu = \mu_s(\xi) + G_j \xi + O(\xi^2)$, where μ_s is the even part of the well and

$$G_j \equiv \partial_x \mu_{\text{out}}(x_j) = \sum_{k \neq j} \left[(1-l)c \sigma_k^I g'_{\ell_I}(x_j - x_k) + d \sigma_k^R g'_{\ell_R}(x_j - x_k) \right] \quad (14)$$

is the gradient of the field due to all *other* spikes. Put $N = N_s(x - X(t)) + \phi$ and linearise:

$$-\dot{X} N'_s = \mathcal{L} \phi + G \xi N_s, \quad \mathcal{L} \equiv D_N \partial_\xi^2 + \mu_s. \quad (15)$$

Here is the structural difference from Gray–Scott. The kernel of $\mathcal{L}$ is N_s itself — (9) says so — and N_s is *even*. Projecting (15) onto it gives $0 = 0$, because $\langle N_s, N'_s \rangle = 0$ and $\langle N_s, \xi N_s \rangle = 0$ by parity. The usual translation-mode solvability is degenerate, and the drift has to be extracted one level down, from the odd mode.

Project instead onto the first excited state ψ_1 , with $\mathcal{L}\psi_1 = -\omega\psi_1$ and $\omega = \sqrt{2D_N\kappa}$ the level spacing:

$$-\dot{X} \langle \psi_1, N'_s \rangle = -\omega \langle \psi_1, \phi \rangle + G \langle \psi_1, \xi N_s \rangle. \quad (16)$$

Defining X as the centroid removes the dipole content of ϕ , so $\langle \psi_1, \phi \rangle = 0$, and for the Gaussian ground state $\psi_1 \propto \xi N_s$ the two remaining brackets are $-\sqrt{\pi}/2$ and $x_0^2 \sqrt{\pi}/2$, giving

$$\boxed{x_j = C x_0^2 G_j, \quad C = O(1), \quad x_0^2 = \sqrt{2D_N/\kappa} \propto \sqrt{D_N}} \quad (17)$$

with $C = 1$ in the harmonic/Gaussian reduction. The measured mobility is $x_0^2 = 4.86 \times 10^{-4}$.

Two things to take from (17). First, the spike climbs the gradient of its environment: it grows faster on the side where μ is larger, and the centroid follows. Second — and this is the key point for the arrest question — the mobility is $O(\sqrt{D_N})$, not $O(D_N)$ and *not* exponentially small in anything. Combined with G being an $O(1)$ function of d/ℓ rather than an exponentially small one, the reduced dynamics has no small parameter left with which to make coarsening take longer than 10^8 .

Remark. The step from (15) to (17) is a two-mode reduction, not a systematic expansion. A systematic version has to work with the full linearised operator including the feedback of ϕ on μ through (2)–(3), whose kernel is N'_s by translation invariance and which is not self-adjoint, so the projection is onto its adjoint null mode. That calculation could change C , and in principle could reveal a mobility much smaller than x_0^2 ; it is the main open point in this document.

6 What this predicts, and where it collides with the data

6.1 Sign and range of the interaction

From (8) the field of one neighbour contributes to μ at weight $(1-l)c\sigma^I$ through the I channel and $d\sigma^R$ through the R channel. With the measured strengths,

$$(1-l)c\sigma^I = -0.01293 \quad (\text{range } \ell_I = 1), \quad d\sigma^R = +0.9304 \quad (\text{range } \ell_R = \sqrt{p}).$$

So I mediates **repulsion** at range ℓ_I (moving towards a neighbour means moving into its depletion hole) and R mediates **attraction** at range ℓ_R (moving towards a neighbour means moving into its byproduct plume). This is the physical content of the p scan: p sets $\ell_R/\ell_I = \sqrt{p}$, i.e. whether the attraction is shorter-ranged than the repulsion.

Balancing the two channels, $(1-l)c|\sigma^I|e^{-s}/2 = \sigma^R e^{-s/\sqrt{p}}/2\sqrt{p}$, gives a crossover distance

$$s^* = \frac{\sqrt{p}}{1 - \sqrt{p}} \log \frac{\sigma^R}{(1-l)c|\sigma^I|\sqrt{p}}, \quad (18)$$

inside which the net force is attractive and outside which it is repulsive — so a lattice with $d > s^*$ is stable and one with $d < s^*$ coarsens, and $s^*(p)$ is the predicted arrest spacing. Holding the $p = 1$ strengths fixed (so this is the shape of the p dependence, not a quantitative prediction away from $p = 1$), $\sigma^R/((1-l)c|\sigma^I|) = 72$ and

p	0.01	0.03	0.1	0.3	1
s^*	0.731	1.263	2.510	5.908	— (no crossover)

At $p = 1$ the ranges coincide, the R channel wins by a factor of 72 at every distance, and there is no stable spacing at all.

6.2 The collision

That last statement is contradicted by the data, and not marginally. Take the arrested state, displace one spike by δ towards its right neighbour, and evaluate (14) and (17) with the strengths held fixed:

δ	G	$v = x_0^2 G$	e -folding time δ/v
0.0122 (1% of d)	$+3.30 \times 10^{-3}$	$+1.61 \times 10^{-6}$	7.6×10^3
0.1221 (10% of d)	$+3.31 \times 10^{-2}$	$+1.61 \times 10^{-5}$	7.6×10^3

The lattice is predicted to be linearly unstable with growth rate 1.3×10^{-4} . The run reached $t = 10^8$ without moving: that is 1.3×10^4 e -folding times. The frozen-strength reduction is wrong by an enormous margin, and the interesting question is which of its ingredients fails. The candidates, in the order I would test them:

1. **The strengths are not frozen** (most likely). σ_k^I and σ_k^R are outputs of the self-consistency in §3, and they depend on the whole configuration. When spike j moves right, its two neighbours acquire *different* masses, and the resulting asymmetry in (14) is a force channel that the frozen calculation discards entirely. In Gierer–Meinhardt and Gray–Scott this is exactly the channel that supplies the stabilising repulsion. Concretely: the positions and the masses must be evolved as one coupled system, not positions in a fixed landscape.
2. **The mobility is much smaller than x_0^2** , because the systematic version of §4 (see the remark) produces a different projection. This would need roughly four orders of magnitude, which seems a lot, but the degeneracy of the leading solvability condition is exactly the kind of structure that produces anomalously small mobilities.
3. **The drift is real but sub-threshold in the diagnostics**. This one is already close to excluded: $v \sim 10^{-5}$ is nine orders above the $\sim 10^{-14}$ noise floor of **peak.drift**, and the run would have merged thousands of times over.

7 The other eigenvalue family

Position modes are only half the spectrum. Perturbing the *masses*, $S_j \rightarrow S_j + \epsilon e^{\lambda t + i q j}$, and solving the outer response produces a nonlocal term $\propto \int \phi$ in the inner eigenvalue problem — a nonlocal eigenvalue problem (NLEP) of the standard type. Its $q \neq 0$ branch is the competition instability: one spike grows at its neighbour’s expense and the neighbour dies. This family is

- $O(1)$ in D_N rather than $O(\sqrt{D_N})$, so it acts first and fast;

- the natural explanation for a *box-independent* arrest spacing, which is what run1 (10 peaks) and run2 (100 peaks) show across a $10\times$ change of L ;
- not obtainable from the position asymptotics at all.

If the arrest is really set by the competition threshold k_c , then the positions can be marginal or even weakly unstable without contradicting the observations — provided the position instability is slow enough, which §6 says it is not, on the frozen-strength estimate. Resolving that tension is the crux.

8 A numerical programme, cheapest first

1. **Displace one spike and measure the drift.** Shift a single spike in the arrested state by 10% of the spacing, re-relax I and R (or just evaluate the residual), and run `peak_drift`. The prediction is $v = +1.6 \times 10^{-5}$ towards the near neighbour. The *sign* says attraction or repulsion, i.e. unstable or stable lattice; the *magnitude* tests (17). Both μ_x and x_0 are measurable directly off the perturbed state, so this tests the drift law with no asymptotics required to evaluate it. This is an afternoon’s work and it discriminates between the three candidate failures above.
2. **Add the virial check (10) to the arrest battery.** Shape-free, cheap, and it cannot be faked by grid pinning.
3. **Solve the inner BVP (13) by Newton.** Gives the $O(1)$ constants and a spike profile to compare with the PDE state.
4. **Build the k -spike quasi-equilibria (§3) and their Floquet spectrum.** Both eigenvalue families at once, for every k , without integrating the PDE. This is the object that answers “is the arrest spacing at the edge of the stable band”.
5. **Only then** the analytic NLEP, if the numerics says the competition threshold is what matters.

9 Caveats

- x_0/ℓ **is not very small.** $x_0/\ell = 0.022$ sounds comfortable, but the effective expansion parameter keeps showing up as something more like $x_0/(D_N/\alpha)^{1/3} = 1.7$ or the 16% correction in (6). Expect $O(1)$ constants to be genuinely $O(1)$, and do not trust a factor of two from this calculation.
- $l = 0.999$ **is a near-singular limit.** $(1-l)cI \approx 10^{-3}$ is negligible in the *value* of μ but its gradient, $(1-l)c \Delta I \approx 2 \times 10^{-3}$, is the same size as μ_0 itself. The I channel cannot be dropped as small; it is small in value and $O(1)$ in effect.
- **No variational structure.** There is no free energy running downhill here, so nothing forces coarsening to proceed, and a stable multi-spike state is not anomalous. The intuition that it must coarsen is imported from Cahn–Hilliard and does not apply.
- **Quasi-staticity of I, R is safe.** The resources relax at $O(1)$ rates against a drift of $O(10^{-5})$, so the adiabatic elimination in §1 is very well satisfied — five orders of margin.

Literature

The closest templates: Doelman, Gardner & Kaper and Kolokolnikov, Ward & Wei for semi-strong interaction and competition/oscillation thresholds in Gray–Scott; Iron, Ward & Wei for the Gierer–Meinhardt spike dynamics and the NLEP machinery; Ei, *The motion of weakly interacting pulses*, for the projection method behind §4; and Sherratt’s Busse-balloon computations for the Klausmeier model, which is the closest structural analogue — slowly diffusing organism plus fast resource, banded patterns, arrested coarsening — and where the stable-wavenumber-band programme has been carried out end to end.

All numbers quoted here are produced by `verify_spikes.jl` in this directory; run it from the repository root with `julia --project=. rest/more_ai_derivations/verify_spikes.jl`.